The Directions of Technical Change

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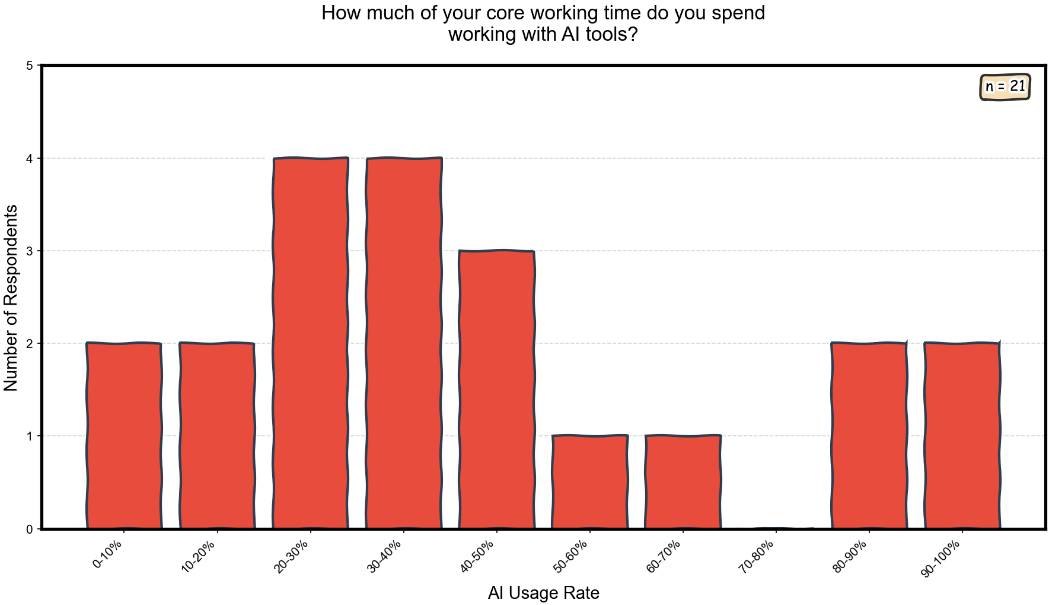
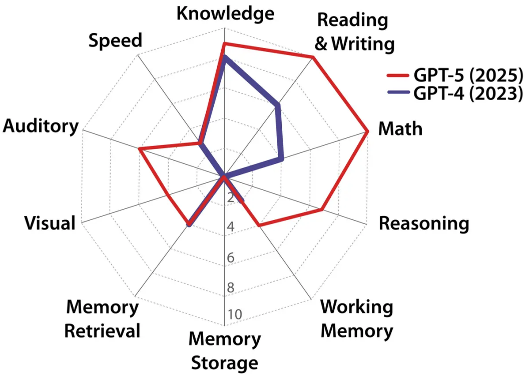
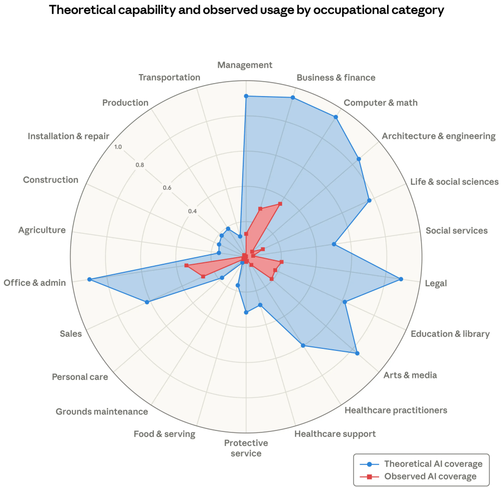
AI adoption at CEU Economics differs widely



AI capability and usage vary widely across tasks



Hendrycks et al. (2025, Fig 1)



Massenkoff and McCrory (2026, Fig 2)

A model of digital technology

1. Technology is **directional** ($t \in \mathbb{R}^N$): excels at some task combinations, not others
2. Adoption is **endogenous**: a collaboration decision under a shared time budget
3. Workers have some **flexibility** in doing their job

Why existing models are insufficient

Too few dimensions

skilled/unskilled, complex/routine, expert $>$ novice

Too rigid

Technology capabilities taken as given. Workers cannot change what they do.

The model

Key ingredients

1. Task bundling: each job is a collection of untradable tasks
2. Job flexibility: workers choose different tasks depending on their skill
3. Substitution patterns driven by spatial proximity

The worker's problem

$$\max_x \; F(x) = \left(\sum_{i=1}^{N} \theta_i^{1/\sigma} x_i^{1-1/\sigma} \right)^{\sigma/(\sigma-1)}$$

subject to

$$\left(\sum_{i=1}^{N} s_i^{-1/\gamma} x_i^{1+1/\gamma} \right)^{\gamma/(\gamma+1)} \leq B$$

x = activities, θ = job requirements (CES), s = skills (CET), B = time in a day.

Two parameters capture job flexibility

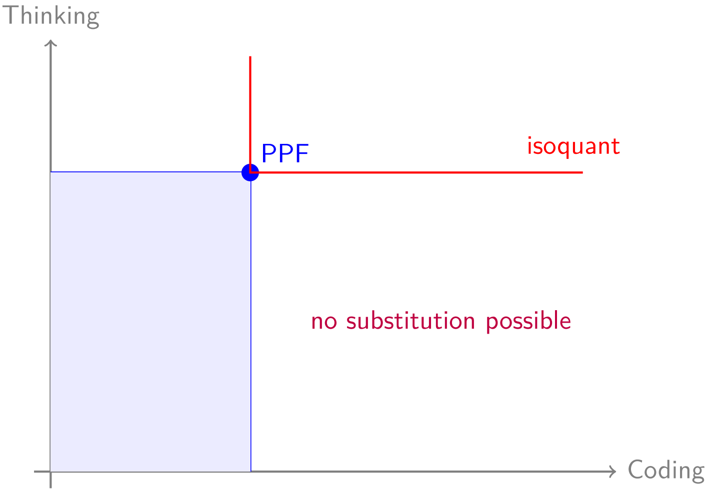
$\gamma =$ elasticity of transformation (PPF curvature)

$\sigma =$ elasticity of substitution (isoquant curvature)

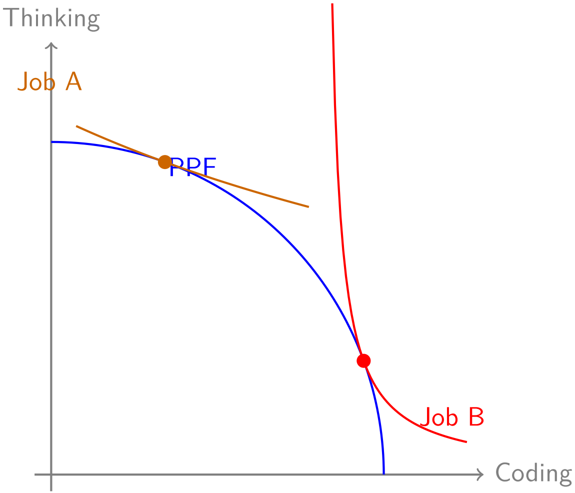
Flexibility

- Knowledge workers choose how to allocate time
- “Shadow IT”: workers use preferred tools regardless of policy
- This model nests the standard case: Leontief $\rightarrow$ employer dictates

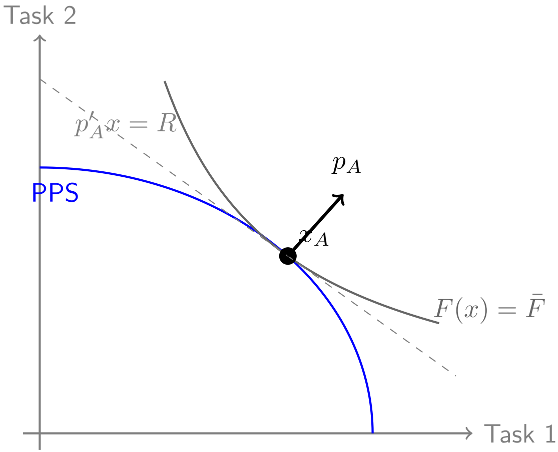
Rigid jobs: no flexibility



Different jobs have different requirements



Autarky prices



At the optimum, the PPF and isoquant are tangent.

The supporting hyperplane defines **autarky (shadow) prices** p^A.

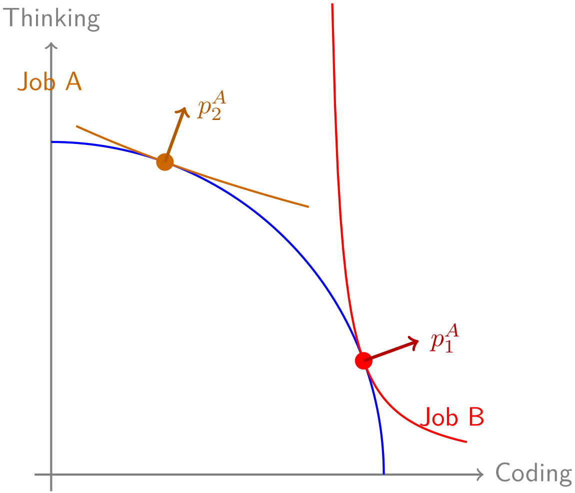
Shadow prices

$$p_i^A \propto \left(\frac{\theta_i}{s_i}\right)^{1/(\gamma+\sigma)}$$

High for activities that are required (θ_i high) but hard to do (s_i low).

Crucially, $p^A \neq x^A$: prices reflect what the worker would like to shift toward, **not** what they currently do.

Autarky prices characterize job activities



Jevons' Paradox

Time share on activity i:

$$\omega_i^A = \frac{s_i^{(\sigma-1)/(\gamma+\sigma)} \theta_i^{(\gamma+1)/(\gamma+\sigma)}}{\sum_j s_j^{(\sigma-1)/(\gamma+\sigma)} \theta_j^{(\gamma+1)/(\gamma+\sigma)}}$$

Increases in skill s_i if $\sigma > 1$. Relevant range for knowledge work.

The general structure

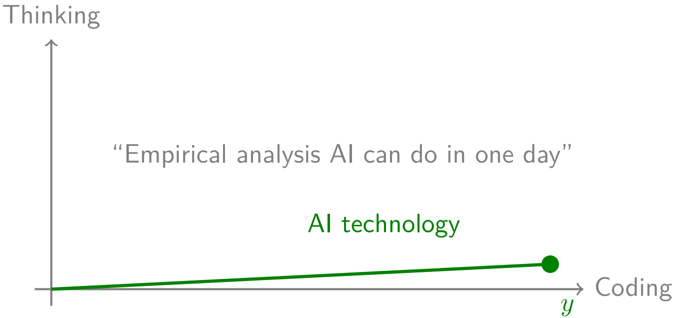
All results depend on **convexity**, not on CES–CET:

- Any convex PPS $X_A = \{x : g(x) \leq B\}$ with quasi-convex hom(1) production function F
- Shadow prices p^A defined by tangency (supporting hyperplane)

CES–CET gives closed forms; the geometry works for any F and g.

Technology

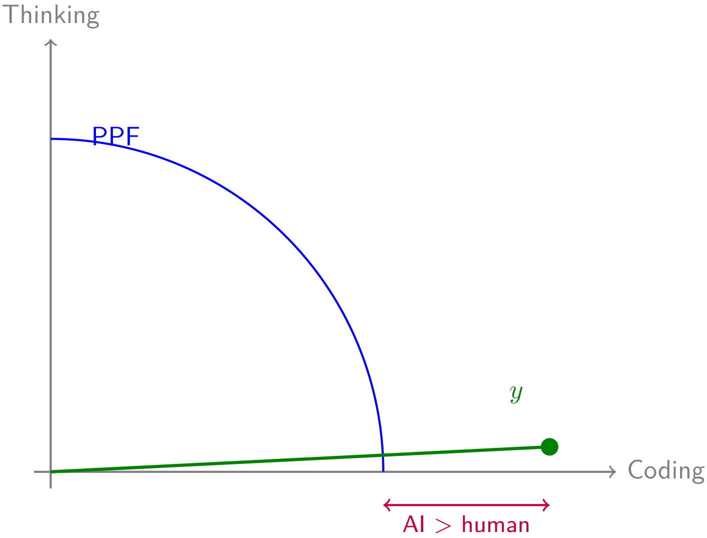
Technology as a spike



Technology (t, χ): direction t (which tasks, $\|t\| = 1$) and capability χ (how much per unit of supervision).

Same resource constraint: one day.

Technology meets the individual



Absolute advantage: $\chi g(t) > 1$. Technology can produce more in direction t than the worker.Necessary but **not sufficient** for adoption.

AI + human = convex hull

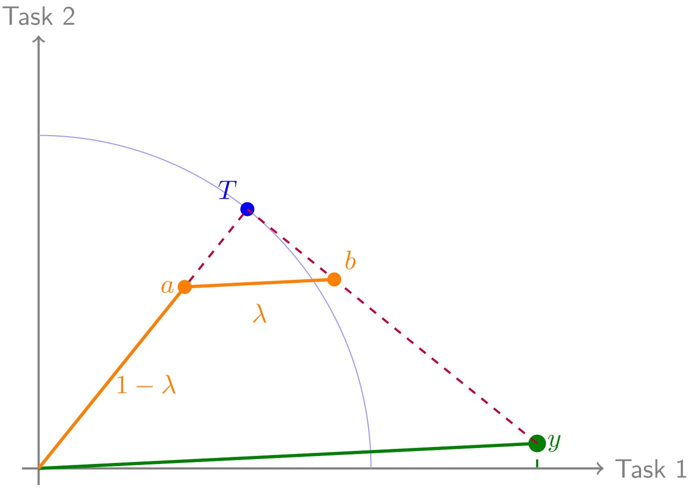
Using technology takes time (giving tasks, supervising, reviewing results).

Spend λ of your day on AI, $1-\lambda$ on your own work.

$$\max_{x,\lambda} \; F\big((1-\lambda)\,x + \lambda\,B\chi t\big) \qquad \text{s.t.} \qquad g(x) \le (1-\lambda)B,\; 0 \le \lambda \le 1$$

The combined PPS is the **convex hull** of X_A and the technology point $B\chi t$.

AI + human = convex hull



$\lambda \in [0,1]$ is a choice variable. May be 0 or 1.

Adoption

The adoption condition

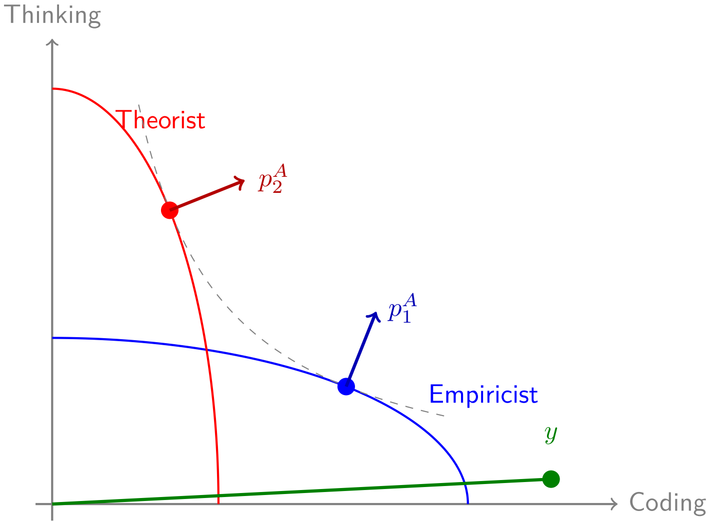
A worker adopts technology (t, χ) if and only if

$$B\chi \cdot p^{A\prime}t > p^{A\prime}x^A = B\varrho(p^A)$$

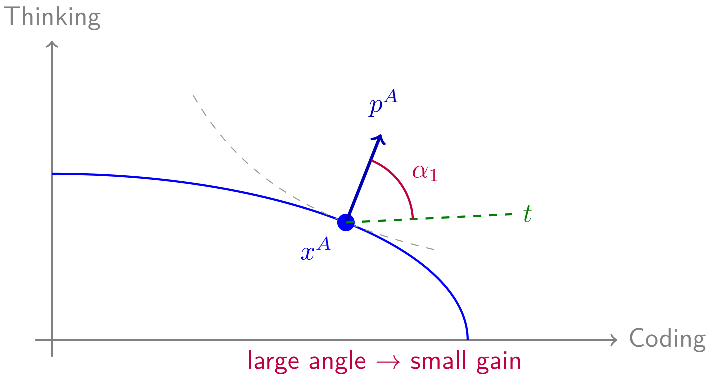
or equivalently: $\chi \geq \underline{\chi}(t, p^A) = \frac{\varrho(p^A)}{p^{A\prime}t}$.

Non-adoption from **standard neoclassical incentives**, not fixed costs.

Two economists



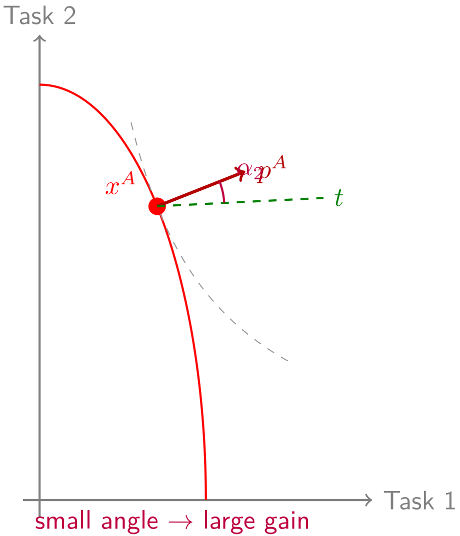
Economist 1: the empiricist



The empiricist's autarky price points toward theory (the expensive activity).

Technology is nearly parallel to what is already **cheap**.

Economist 2: the theorist



The theorist's autarky price points toward empirics (the expensive activity).

Technology is nearly parallel to what is **expensive**: big gain.

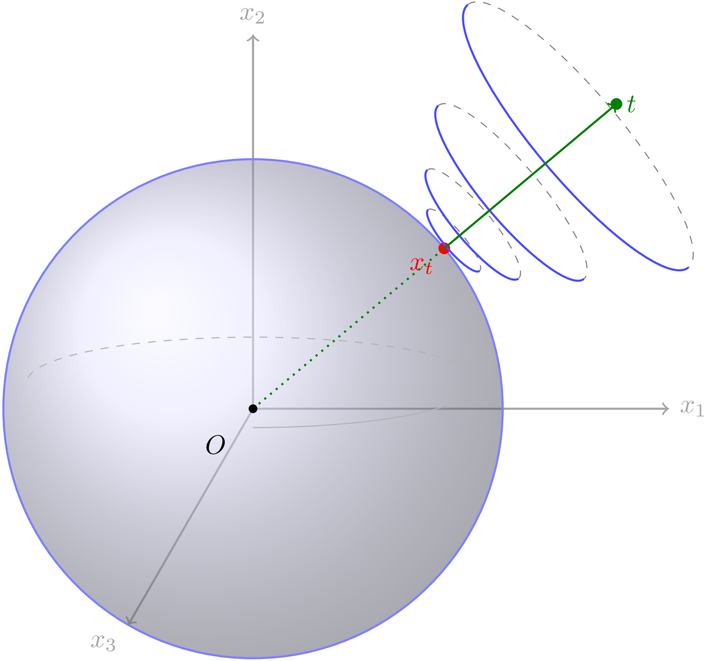
The alignment angle determines everything

$$\text{Adoption iff } \chi \cos\phi \geq \varrho(p^A)$$

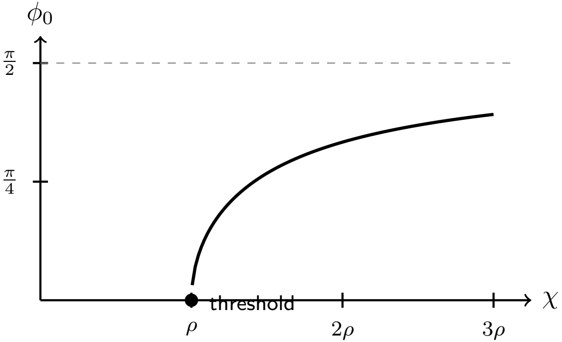
where $\phi = \arccos(p^{A\prime} t)$ is the **alignment angle**.

- Small $\phi \to$ low bar (technology targets your weakness)
- Large $\phi \to$ high bar (technology targets your strength)

The cone of technology adoption



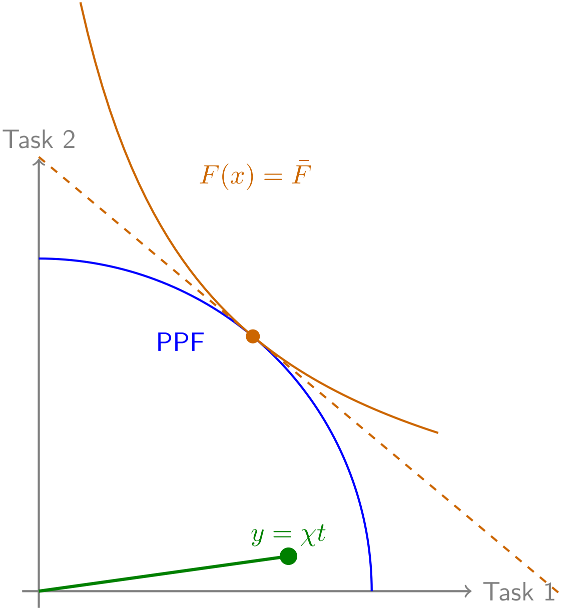
Suddent adoption



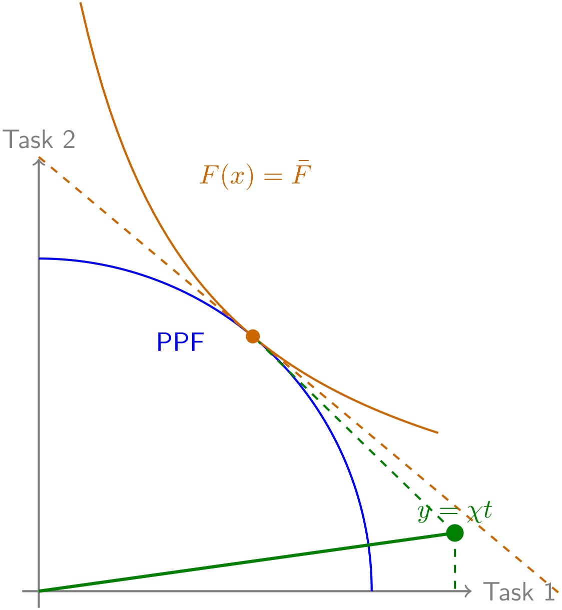
Near the threshold, $\phi_0 \approx \sqrt{2(\chi/\varrho - 1)}$: **steep S-curves** in adoption.

The intensive margin

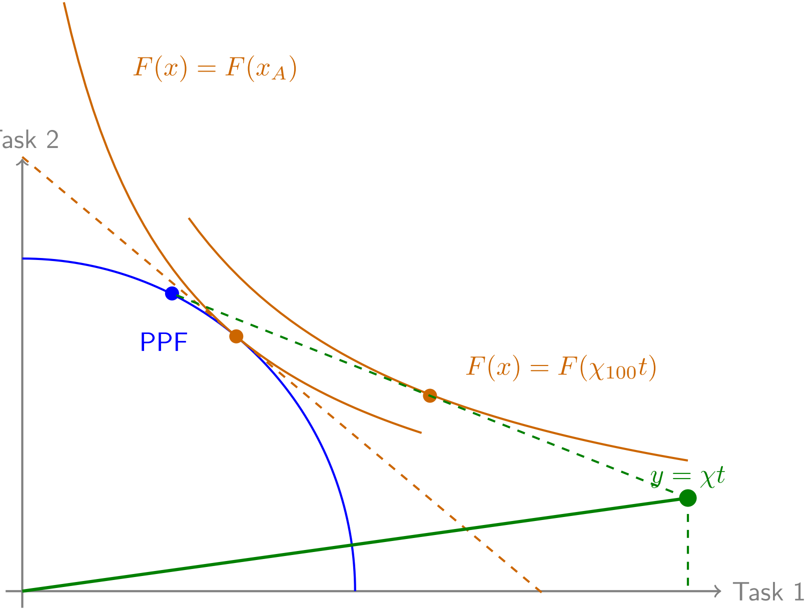
AI has no absolute advantage in anything



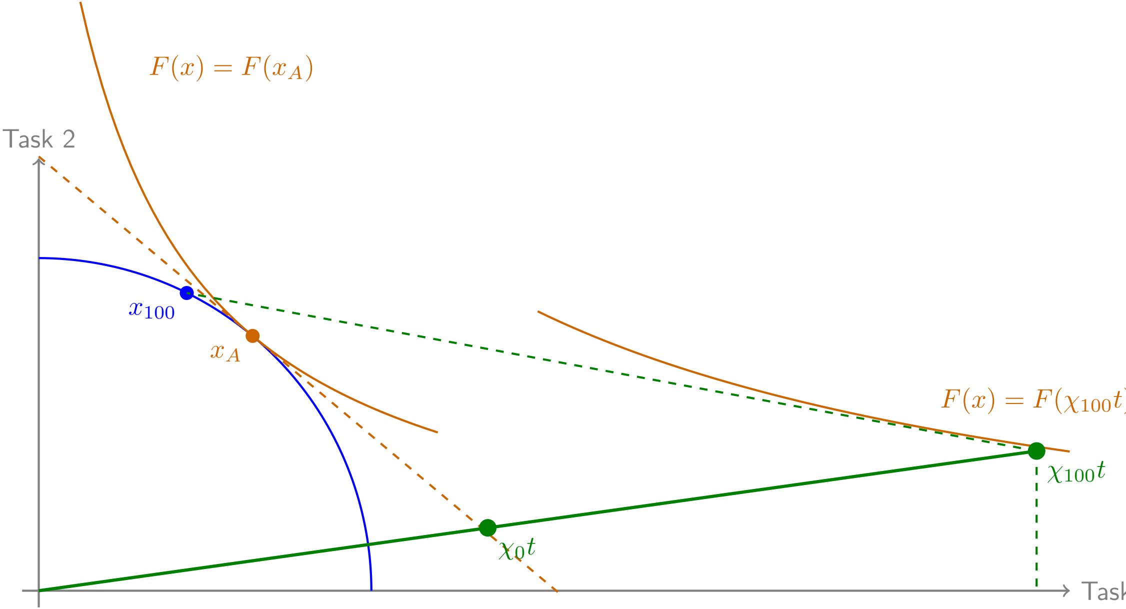
AI has an absolute, but not comparative advantage



Worker starts adopting AI



Worker specializes in their comparative advantage



Two thresholds

Entry threshold: $\chi_0 = \underline{\chi}(t, p^A)$ — worker starts using AI.

All-in threshold: $\chi_{100} = \underline{\chi}(t, p_{100})$ — worker uses AI 100% of the time.

For $\chi \in (\chi_0, \chi_{100})$: **partial adoption** with $\lambda^* \in (0, 1)$.

Gap $\chi_{100} - \chi_0$ **narrows** with flexibility (γ, σ): flexible jobs adopt faster but also go all-in sooner.

As technology capability increases

1. Absolute advantage is necessary but **not sufficient** for adoption
2. Rapid adoption after the threshold (steep S-curves near entry)

- adoption is worker-job specific

3. Structured intensive margin: partial $\rightarrow$ all-in

- Work activities become *less similar* across workers who adopt
- Output becomes *more similar*

5. Some people never adopt $(N > K)$

Summary

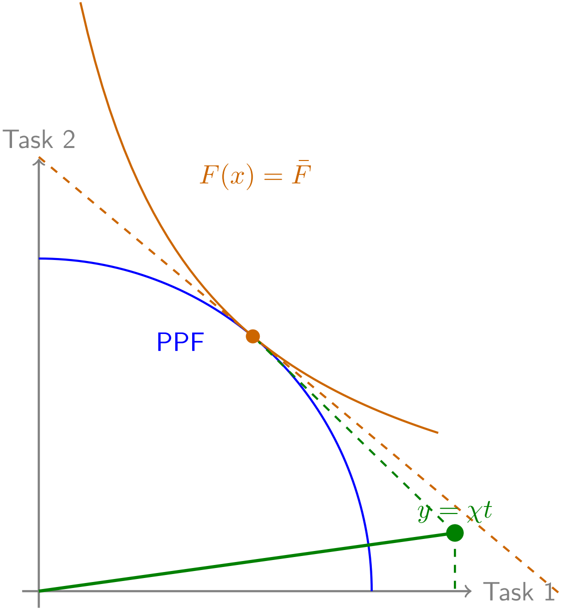
AI adoption is a **team-production** decision under a shared time budget.

Autarky prices — not skills, not job requirements, not activities — are the sufficient statistic.

A tool that outperforms the worker need not be adopted if those tasks are not the binding constraint.

As capability improves, adoption expands in a **cone** centered on autarky prices, with steep S-curves near the threshold.

The figure to take away



Appendix

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